

FFGFT — Epistemological Position

4D Description, Natural Units, and Self-Referentiality

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This document summarises the epistemological position of FFGFT. It addresses: (1) the 4D description (fractal, open, recursive), (2) equivalence of all descriptions in natural units, (3) structure of the vacuum and T-field, (4) FFGFT position on black holes, (5) self-referentiality of any inside description, and (6) why T^4 as a posit is operationally superior to “pure information”. A closing remark draws the parallel to the Logos concept.

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1. The 4D Description Is Fractal, Open, and Recursive

FFGFT works with the 4D identification torus structure T^4 — not a closed, deterministic system.

Fractal: $D_f = 2.999867$ — subtle but physically consequential.

Open: Why T^4 , why $\xi = 1.333 \times 10^{-4}$ — remains explicitly open. No ontological completeness claim.

Recursive: ξ -ladder with natural cut-off $N \approx 8-9$, where Z_3^3 topology and Fibonacci convergence coincide.

2. The 4D Description Does Not Apply to the Totality

FFGFT does *not* claim that the entire domain of existence is described by 4D. Derivation framework: *given T^4 and ξ , everything else follows*. What lies before T^4 remains open.

3. The Lower Bound: 4D Applies to the Absolutely Smallest

$$L_0 = \xi \cdot \square_P \approx 2.2 \times 10^{-39} \text{ m}$$

Below L_0 , T^4 has no well-defined degrees of freedom. Boundary applies *downward*. Torus size is an emergent coherence length, not a fixed parameter.

4. Natural Units: No Privileged Description

$$E \leftrightarrow m \leftrightarrow 1/t \leftrightarrow f$$

$E = mc^2$, $E = hf$, $T \cdot m = 1$ (T = T-field operator, FFGFT's foundational relation), $f = 1/t$. None is privileged — all are projections of the same geometric reality on T^4 . [Pascher 2026]

5. Consequence: Information vs. Geometry

“Information” as a description is a valid projection — but not more fundamental than the geometric description.

Information is distinction. Shannon (1948), Bateson (1972), IPI 2026: without distinction, there is no information. “Pure unrestrained information” without structure is indistinguishable from nothing — not a primitive, but the absence of everything.

Consequence for Atticus's Postulate: Continuous spin, unbounded velocity, no quantum numbers — this is not “pure information” but **the absence of all structure through which information would be defined**. Not connectable to any known or coherently conceivable physics.

6. The Vacuum Has Structure — No Field Without Structure

6.1. The Vacuum Is Not Empty

ξ -field ground state: vacuum energy density as geometric property, $T \cdot m = 1$ active in the ground state, $L_0 = \xi \cdot \ell_P$ as structural lower bound.

6.1.1 Thermodynamic Equilibrium Changes Nothing

Even in total equilibrium, the field remains structured: field equations hold, T^4 topology exists, ξ retains its value.

Equilibrium is not the absence of structure. Without field structure there is no concept of equilibrium.

6.2. FFGFT Is a Field Theory — No Field Without Structure

No field without structure — mathematical tautology. “Pure unstructured information” cannot carry a field, build a vacuum, or satisfy a field equation.

7. Black Holes — Horizon, Singularity, and Fundamental Information

7.1. Black Holes Are Points in the Universe, Not the Universe

Event horizon and cosmological horizon are **epistemic** boundaries — not ontological statements. We can only describe what we see.

7.2. Black Holes Are Not Singular in FFGFT

$L_0 = \xi \cdot \ell_P$ is an absolute lower bound — infinite density is structurally excluded. $T \cdot m = 1$ remains valid. Black holes are T^4 winding configurations. [Pascher 2026]

7.3. FFGFT Does Not Lose Fundamental Information

Winding numbers (n_θ , n_ϕ , n_3 , n_4) are *topological invariants*. Information loss is an **access problem** (epistemic), not an **existence problem** (ontological). [Bekenstein 1973; Hawking 1974]

8. Self-Referentiality — We Always Describe From the Inside

8.1. We Are Part of the System

Neurons, electrochemical signals, conscious processes — all within T^4 . The description is a process *within* the system acting as if it stands above it.

We are part of the system. We always describe from the inside, never from the outside.

8.2. The Thought Experiment of an Outside View

The attempt to adopt an outside view is itself a material process within T^4 . Every apparent outside view is an inside view dressed as an outside view.

8.3. The Recursive Situation

Matter → Description → Matter → Description → ...

In FFGFT, this recursion does not terminate — consistent with its fractal, open, recursive character.

Gödel Analogy [1931]: A sufficiently powerful formal system cannot prove its own consistency from its own axioms. “Why T^4 ?” remains explicitly open — structural honesty, not a gap.

Why T^4 and Not “Pure Information”? T^4 is **operational**: derivations (winding numbers, ξ , L_0 , field equations), measurable predictions. “Pure information” is **not operational**: no derivation, no distinction possible, coherence already presupposes structure. T^4 proves itself. “Pure information” does not.

8.4. Consequence for the Postulate of Pure Information

“Pure information before geometry” implicitly assumes an outside view — a standpoint beyond all structure. But this standpoint is itself a material process within T^4 . FFGFT accepts this limit.

Closing Remark: “In the Beginning Was the Word”

The discussion about the primitive leads, in principle, consciously or unconsciously, to: “In the beginning was the Word” (John 1:1). The Logos concept is philosophically precise: the Word is structure, order, distinction — exactly what information in the Shannon sense presupposes.

The postulate of “pure, unrestrained information” and the Johannine Logos share the same structural problem: both require someone *outside the system* to speak. The writers of the Bible were human — material beings within the system.

The difference in one sentence: The Bible claims an outside view it cannot have. FFGFT relinquishes this claim and names the boundary explicitly.

Summary

Claim	FFGFT Position
4D is the only possible description	No — 4D is the framework, ontologically open
4D applies to the totality of existence	No — FFGFT makes no total claim
4D applies to the absolutely smallest	Yes — $L_0 = \xi \cdot l_P$ sets the lower bound
Below L_0 nothing more exists (in FFGFT)	Yes — no stable degrees of freedom

Claim	FFGFT Position
Geometry is privileged over energy / mass / time	No — all projections are equivalent
Unstructured information as primitive	Contradiction — information is distinction, hence structure
Atticus's postulate is physically connectable	No — empty name, not operational, no field equation possible
The vacuum is structureless	No — ξ -field ground state, geometrically structured
A field can exist without structure	No — mathematical tautology
Black holes are singular in FFGFT	No — L_0 structurally prevents infinite density
Black holes destroy information	No — topological invariants are preserved
Horizons are ontological boundaries	No — epistemic limits of the observable
We describe FFGFT from outside	No — material T^4 processes describing from within
The recursion matter→description terminates	No — open recursion, consistent with FFGFT's character
T^4 and “pure information” are equivalent posits	No — T^4 is operational; “pure information” has no derivation power
“Why T^4 ?” is answered	No — structural honesty, not a gap
The Bible adopts an outside view	Claim from inside — FFGFT names the boundary explicitly

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